Hamze Dokoohaki

AI Systems Architect & Technical Leader

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AI systems architect who builds production-grade intelligent systems at the intersection of deterministic control and model capabilities. Designed and shipped a **20+ microservice voice AI platform** handling real-time multi-agent orchestration, built **RL-driven prompt optimization systems**, and implemented **AI safety infrastructure** from the ground up. Background spans ecosystem modeling, physics-informed neural networks, and healthcare AI—the common thread is building systems where AI is powerful where it needs to be and predictable where it must be.

TECHNICAL PHILOSOPHY

State Machines + LLMs

LLMs are powerful but unpredictable—they shouldn't orchestrate. I build explicit state machines as the control plane, with LLMs handling NLU/NLG within bounded contexts.

Safety as Infrastructure

Safety isn't a filter—it's architectural. Input sanitization, prompt injection defense, PII detection, and human handoff built into every layer.

Evaluation-Driven Development

Automated evaluation infrastructure (agent gyms, LLM-as-Judge, voice benchmarks) closes the loop between deployment and optimization.

KEY SYSTEMS BUILT

Production Voice AI Platform

Blooming Health

Python | FastAPI | LangGraph | WebRTC | Twilio | Redis | PostgreSQL | Docker | CUDA

Production-grade voice AI platform—**20+ microservice system** orchestrating real-time, multi-turn conversational agents across voice (WebRTC, PSTN), SMS, and text. A full operating system for conversational AI.

- **State Machine as Control Plane:** 2,600-line deterministic state machine governing multi-agent conversations—LLMs handle language within bounded contexts, but routing/transitions/safety follow explicit state logic
- **Composable Conversation Graphs:** 10 workflow processors as traversable state graphs with cursor navigation and conditional edges—complex interactions become predictable paths
- **Hot-Swapping Agent Context:** Mid-conversation context switching without dropping calls—VAD gating for state synchronization
- **Automated Prompt Optimization:** RL system analyzing failures and generating improved variants (textual gradient, evolutionary, Bayesian strategies)
- **Continuous Agent Evaluation:** Gymnasium-compatible framework running **10-1,000 parallel episodes** with LLM-powered simulators
- **Production Audio Pipeline:** 4 STT engines, 4 TTS engines, 4 noise suppression algorithms, dual-trigger VAD for natural turn-taking

Kashef.ai - Knowledge Discovery Platform

Kashef AI LLC (Founder)

Python | LangChain | pgvector | Knowledge Graphs | RAG | Multi-Agent

Multi-modal knowledge discovery combining semantic embeddings, heterogeneous knowledge graphs, and agentic orchestration for cross-domain pattern discovery.

- **Beyond Vector Search:** Embeddings + graph traversal (centrality, community detection) to surface multi-hop insights pure RAG misses
- **Composable Agent Capabilities:** Plugin architecture where skills register with metadata/schemas—agents compose at runtime
- **Defensible AI Outputs:** DAG-based provenance so every insight traces to source documents

Physics-Informed Neural Network Platform

Indigo Ag

PyTorch | AWS SageMaker | Differentiable Programming | Bayesian Methods

Neural networks learning residual corrections to physics-based models—accuracy of deep learning with physical consistency. Processing **300GB+ geospatial data**, **20% improvement** over traditional models. **Provisional patent** for novel differentiable coupling architecture.

AI Knowledge Synthesis System

Indigo Ag

Local LLMs | RAG | React | PostgreSQL

Award-winning tool automating literature analysis across **3,000+ papers**. Multilingual discovery surfaces non-English research; traceable extraction links every insight to source passages. Two-time hackathon winner with demonstrated commercial value.

CORE COMPETENCIES

LLM Engineering
Multi-provider (OpenAI, Anthropic, Gemini)
LangChain/LangGraph orchestration
MCP for tool execution
Production prompt engineering
OpenAI Realtime API

Multi-Agent Systems
Deterministic state machines
Agent handoff patterns
Tool execution governance
Skill/capability registries
Human-in-the-loop design

AI Safety
Prompt injection defense
PII detection & filtering
Output safety monitoring
Risk tracking & escalation
Rate limiting & abuse prevention

Evaluation & Optimization
Gymnasium agent frameworks
LLM-as-Judge scoring
RL prompt optimization
Voice metrics (WER, CER)
Automated A/B testing

Speech & Audio AI
STT (Whisper, Deepgram, FunASR)
TTS (OpenVoice, Cartesia, MeloTTS)
VAD (PyAnnote, Silero, semantic)
Noise suppression (DTLN, RNNoise)
Real-time audio pipelines

Infrastructure
Docker/Compose (20+ services)
GPU inference (BentoML, ONNX)
CI/CD (GitHub Actions, GCP)
Redis, PostgreSQL, pgvector
WebRTC/Twilio real-time

EXPERIENCE

Blooming Health
Oct 2025 - Present
New York, NY

Director of AI

> Leading AI strategy and architecture for healthcare technology platform
> Built production voice AI system from ground up (20+ microservices)
> Designed state machine orchestration ensuring deterministic control over multi-agent conversations
> Hiring and building AI engineering team

Kashef AI LLC
2025 - Present
New York, NY

Founder

> Founded AI consulting firm for knowledge discovery and agentic systems
> Built platform combining semantic search, knowledge graphs, and multi-agent orchestration

Indigo Ag
Feb 2023 - Present
Boston, MA (Remote)

Principal Data Scientist → Technical Consultant

> Operationalized physics-informed neural network; secured provisional patent
> Built award-winning AI knowledge synthesis platform (3K+ papers)
> Contributed to largest agricultural soil carbon deal ($2.85M Microsoft credits)

U. Illinois Urbana-Champaign
Jun 2020 - Present
Urbana-Champaign, IL

Assistant Professor → Adjunct Professor

> Led digital agriculture research: data fusion, remote sensing, predictive modeling
> PI/Co-PI on $4.6M+ grants; 25+ peer-reviewed publications

Boston University
Jun 2018 - May 2020
Boston, MA

Postdoctoral Research Associate

> Built continental-scale data assimilation for carbon cycle MRV

EDUCATION

Ph.D. Crop Production & Physiology
Iowa State University
2018

M.Sc. Irrigation & Drainage
Isfahan University of Technology
2013

B.Sc. Water Engineering
Shiraz University
2010

Publications & IP: 25+ peer-reviewed publications (Science of The Total Environment, Geoscientific Model Development, Remote Sensing). Provisional patent for Physics-Informed Neural Network architecture.